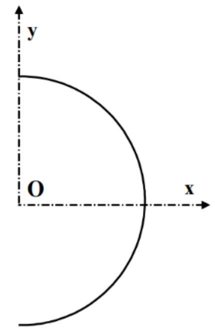
Geometry of Mass

Exercise 01: Demi-circumference

Let there be a semicircle of radius R, centered at O and with linear mass.

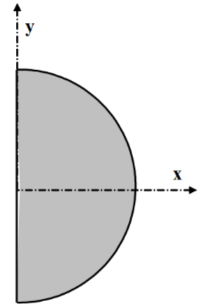
Determine the position of the center of gravity G.



Exercise 02: Half-disk

Let there be a semicircle of radius R, with center O and surface mass.

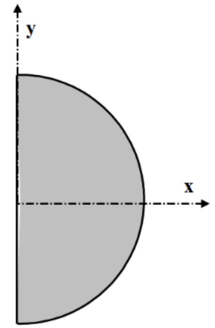
Determine the position of the center of gravity G.



Exercise 03: Hemisphere

Let there be a hemisphere of radius R, centered at O and with a density.

Determine the position of the center of gravity G.

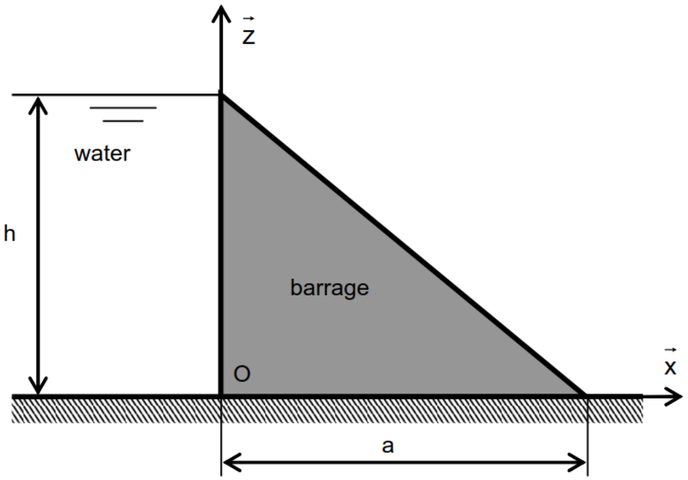


Exercise 04: Weight barrage

A concrete gravity dam, with a triangular cross-section, rests on the ground and creates a water reservoir of height h.

NB: It is located in the middle of the dam in the width direction (following y).

1- Give the surface S of a section of the dam. Retrieve this result by integrating a small element of surface: $S = \int_s \; ds$
2- Determine the position of the center of gravity G.



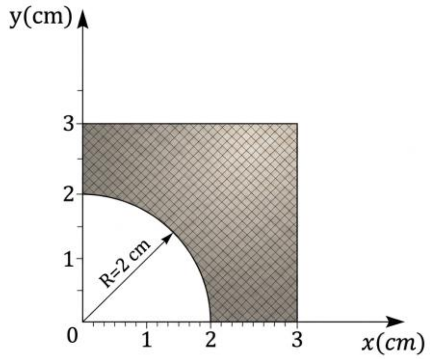
Exercise 04: Guldin's theorem

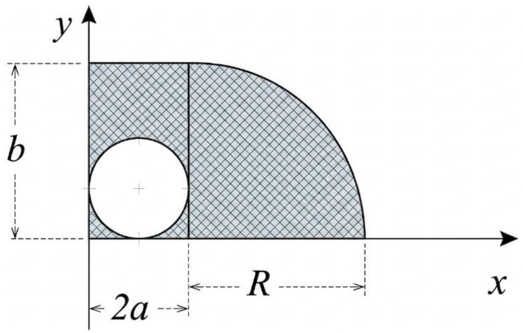
Q1. Find the result of 1he exercise (semicircle) by the Guldin's theorem.

Q2. Find the result of 2th exercise (half-disk) by the Guldin's theorem.

Exercise 05:

Determine, by integration and Guldin's theorem, the coordinates of the centroids of the following homogeneous planar bodies.





Exercise 06:

A square plate of side a and mass m is cut out from a solid, homogeneous disk of mass M and radius R, as shown in the figure.

1. **Determine** the inertia tensor of the disk with respect to the coordinate frame $R_0\ (O, \vec{x_0}, \vec{y_0}, \vec{z_0})$
2. **Determine** the inertia tensor of the plate in frame $R_1\ (O, \vec{x_1}, \vec{y_1}, \vec{z_1})$, then in frame $R_0\ (O, \vec{x_0}, \vec{y_0}, \vec{z_0})$
3. **Deduce** the inertia tensor of the system in frame $R_0\ (O, \vec{x_0}, \vec{y_0}, \vec{z_0})$.